

DATS6401 Final Term Project — Presentation Script

Jeongwon Yoo | Online Shopping Analytics Dashboard

Total Time: 20 minutes

발표 시작 전 체크리스트

- ☐ 브라우저에 <http://127.0.0.1:8050> 열어놓기
 - ☐ PyCharm에서 app.py 실행 중인지 확인
 - ☐ PPT 열어놓기
 - ☐ 마이크 테스트
-

SLIDE 1 — Title (30초)

"Good [morning / afternoon], everyone.

My name is Jeongwon Yoo, and today I'll be presenting
my Final Term Project for DATS6401 —
Visualization of Complex Data.

The project is called the Online Shopping Analytics Dashboard —
an interactive, web-based application built with Python and Dash,
designed to explore and visualize a real-world e-commerce dataset.

Let's get started."

SLIDE 2 — Agenda (30초)

"Here's what I'll cover today.

I'll start with a quick overview of the dataset and key findings,
then walk through the statistical analysis I performed.
After that, I'll do a live demo of the interactive dashboard,
and wrap up with conclusions and business recommendations.

Let's dive in."

SLIDE 3 — Dataset Overview (1분 30초)

"The dataset I selected is an Online Shopping Transactions dataset.

It contains 52,924 transaction records with 23 features.

covering purchases made in 2019 across five US cities —
Chicago, California, New York, New Jersey, and Washington DC.

There are 1,468 unique customers and over 20 product categories.

For this project, I defined Revenue —
calculated as Quantity times Average Price —
as my dependent variable.

The independent variables include customer demographics
like Gender and Location, transactional features like
Avg Price and GST, and behavioral features like
Online and Offline Spending.

As you can see from the pie chart,
62% of customers are female —
a finding that has direct implications
for targeted marketing strategy."

SLIDE 4 — Key Findings (2분)

"Let me highlight the six most interesting findings
from my analysis.

First — Apparel is the top product category
with over 18,000 transactions,
and Chicago is the most active city.

Second — and this was the most surprising finding —
GST shows a strong NEGATIVE correlation
with Average Price, at $r = \text{negative } 0.58$.
This means lower-priced items carry higher tax rates —
revealing a tiered tax structure.

Third — Online Spend follows a near-uniform distribution
between \$320 and \$4,557, which is unusual.
This suggests customers spend at consistent tiers,
possibly due to subscription or bundle packages.

Fourth — there's a clear seasonal peak
in months 8 through 10 — August to October —
where revenue spikes significantly.

Fifth — coupon conversion gap:
50.88% of customers clicked coupons
but only 33.83% actually redeemed them.

That's a 17% gap — a real business opportunity

that's a 17% gap — a real business opportunity.
And sixth — Offline spending at \$2,831 average consistently outperforms Online at \$1,893 — meaning an omni-channel strategy is essential."

SLIDE 5 — Statistical Analysis (2분)

"Moving into the statistical analysis.

I ran normality tests on all numerical features using both Shapiro-Wilk and D'Agostino K-squared tests.

As you can see in the table, every single feature returned a p-value below 0.001 — meaning none of them follow a normal distribution.

This is expected for transaction data, which tends to be right-skewed.

Because of this, I applied log transformation, which significantly reduced skewness and brought the distributions closer to normal.

On the right, you can see my PCA results. PC1 explains 24.2% of the variance and is dominated by spending-related features. Just 5 principal components capture 77% of total variance — meaning we can significantly reduce dimensionality without losing much information."

SLIDE 6 — Dashboard Demo Intro (30초)

"Now let me show you the dashboard I built.

It's organized into 9 interactive tabs, each covering a different analytical function.

The key thing I want you to notice is — every graph updates in real time without re-executing the code.

Let me switch to the browser now."

LIVE DEMO (8분)

TAB 1 — Load Data (40초)

"Here's the dashboard homepage — the Load Data tab.

[행 수를 10 → 20으로 바꾸고 Load Data 클릭]

You can see it immediately updates the table.

I can also filter by any column using this dropdown.

The badges at the top give an instant summary —
52,924 rows, 23 columns, 800 missing values,
and 1,468 unique customers.

Below we have the full descriptive statistics table
for all numerical features."

TAB 2 — Data Cleaning (40초)

"Next is the Data Cleaning tab.

[RadioItems에서 Drop NA → Fill with Mean으로 바꾸고 Apply 클릭]

I implemented five cleaning methods —
Drop, Mean fill, Median fill, Mode fill, and Forward Fill.

The LED displays on the right show
missing values before and after —
for Online_Spend, we had 31 missing values
which are now reduced to zero.

[다시 Drop NA로 바꾸기]

The histograms update instantly to confirm
the distribution shape is preserved after cleaning."

TAB 3 — Outlier Detection (50초)

"The Outlier Detection tab.

[IQR 선택된 상태에서 Detect Outliers 클릭]

Three detection methods are available —
IQR, Z-Score, and Modified Z-Score.

The circular gauge on the right shows

the circular gauge on the right shows the outlier percentage in color-coded zones — green for under 5%, yellow for 5-15%, red above 15%.

[Slider를 1.5 → 2.0으로 조정하고 다시 클릭]

When I change the IQR multiplier to 2.0, you can see the outlier percentage drops and the gauge updates immediately.

For Online_Spend with default IQR, 265 outliers were detected — just 0.50%, indicating a relatively clean distribution."

TAB 4 — PCA (50초)

"Principal Component Analysis tab.

[Run PCA 클릭]

The scree plot shows each component's explained variance percentage.

PC1 explains 24.2% and is dominated by spending-related features.

[Color by 드롭다운을 Gender → Location으로 바꾸고 Run PCA 다시 클릭]

I can change the coloring to Location — and the biplot updates to show how customer data clusters by city.

The GraduatedBar on the top right shows cumulative variance for the selected components — currently 77% with 5 components.

The condition number is very large — about 589 trillion — which indicates multicollinearity between derived features like Total_Spend and Revenue."

TAB 5 — Normality Test (50초)

"Normality Test tab.

[Online_Spend 선택 → Both 선택 → Run Test 클릭]

Both Shapiro-Wilk and D'Agostino K-squared tests confirm Online_Spend is NOT normally distributed —

p-value is essentially zero.

You can see the DAQ components here —
the blue thermometer shows skewness at 0.45,
the red thermometer shows kurtosis at negative 0.15,
and the boolean switch shows 'Not Normal.'

The knob on the right displays the p-value.

[ColorPicker로 색상 바꾸기]

I can also change the plot color using this color picker —
and the histogram updates instantly."

TAB 6 — Data Transformation (40초)

"Data Transformation tab.

[Log 선택된 상태에서 Apply 클릭]

Applying log transformation to Online_Spend —
you can see the skewness changes
from 0.45 to negative 0.60 in the LED displays,
and the histogram on the right
closely approximates the normal PDF curve now.

[Box-Cox로 바꾸고 Apply 클릭]

Switching to Box-Cox transformation —
different method, different result.
All happening without touching the code.

[Download CSV 버튼 보여주기]

And I can download the transformed dataset as CSV
directly from the dashboard."

TAB 7 — Numerical Plots (1분)

"Numerical Plots tab — this one has 13 plot types.

[DropDownMenu 클릭해서 열기]

I can select any plot type from this dropdown menu.

[Histogram + KDE 선택 → Plot 클릭]

Starting with Histogram + KDE for Online_Spend,
colored by Gender.

You can see the multimodal distribution
and female customers dominating every peak.

[RangeSlider 조정]

The range slider controls the bin count —
let me increase it for more detail.

[Scatter + Regression 선택 → X: Avg_Price, Y: GST → Plot 클릭]

Now switching to Scatter with Regression —
setting X to Avg_Price and Y to GST —
this is the key finding: $r = \text{negative } 0.58$.
The regression line clearly shows
higher prices associate with lower tax rates.

[3D Scatter 선택 → Plot 클릭]

And here's the 3D Scatter plot —
fully interactive, you can rotate it."

TAB 8 — Categorical Plots (40초)

"Categorical Plots tab.

[Product_Category 선택, Bar Plot sorted, Plot 클릭]

The sorted bar chart shows
Nest generates the highest mean revenue at \$235,
followed by Nest-Canada and Nest-USA.

[Pie Chart 선택 → Gender → Plot 클릭]

Switching to Pie Chart for Gender —
62% female, 38% male, instantly.

[Stacked Bar 선택, Location, hue Gender → Plot 클릭]

And stacked bar by Location and Gender —
Chicago clearly leads with 18,380 transactions."

TAB 9 — Statistics (40초)

"Finally, the Statistics tab.

[Update 클릭 — Heatmap]

The Pearson correlation heatmap —
I can change which features to include

using the checklist on the left,
and change the color scale from this dropdown.

The strongest correlation highlighted:
Avg_Price and GST at negative 0.58.

[Update — Pair Plot with Gender]

The scatter matrix colored by Gender
confirms male and female customers
show nearly identical spending distributions.

[Update KDE — Online vs Offline]

And the multivariate KDE shows
the joint density of Online vs Offline Spend —
the brightest region is where most customers sit:
Offline around \$2,500-3,000,
Online around \$1,500-2,000."

 브라우저 → PPT로 다시 전환

SLIDE 7 — Correlation Highlights (30초)

"As shown in the demo,
the strongest correlation with Revenue is Total_Spend
— and the most interesting finding remains
the negative GST-price relationship.

Online spending averages \$1,893
while Offline averages \$2,831 —
confirming the continued importance
of physical retail."

SLIDE 8 — Conclusions (1분)

"To wrap up — six key takeaways.

One: Target female customers with personalized

Apparel and Nest-USA campaigns.

Two: Maximize promotions before August
to capitalize on the seasonal peak.

Three: Review the tax structure for low-price, high-GST items.

Four: Improve coupon redemption flow to close the 17% conversion gap.

Five: Invest in offline experience — it consistently outperforms online.

And six: The dashboard enables non-technical stakeholders to explore data interactively, without writing a single line of code. That's the real value of what I've built.

Thank you — I'm happy to take any questions."

? Q&A — 예상 질문 & 답변

Q: Why did you choose this dataset?

"It covers both online and offline spending behavior, which allows for rich multi-channel analysis.

The combination of transaction data, demographic info, and coupon usage made it ideal for visualization."

Q: Why Dash over Streamlit?

"The course required Dash with dash-bootstrap-components.

Additionally, Dash offers more fine-grained control over layout and callbacks,

which was important for the level of interactivity required — especially the DAQ components like the Gauge and Thermometer."

Q: What was the most challenging part?

"The most technically challenging part was handling the color parameter issue in Plotly Express — when the hue was set to None,

it caused an `AttributeError` on the constructor.

I fixed it by switching from `px.histogram` with marginal KDE to a custom `go.Histogram` plus manual KDE overlay."

Q: What does the condition number tell us?

"The condition number of 589 trillion indicates severe multicollinearity. This is expected because Total_Spend is literally the sum of Online and Offline Spend, and Revenue is Quantity times Avg_Price. So the features are not independent by design."

Q: What would you add with more time?

"I would add a time-series forecasting tab using Prophet or ARIMA to predict future revenue, and a customer segmentation tab using K-means clustering on PCA-reduced features."

스크립트 끝 — 총 예상 시간: 약 20분